

Civics Group Reg Number

Candidate's Name _____

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MERIDIAN JUNIOR COLLEGE

JC2 Preliminary Examinations

Higher 1

H1 Biology

Paper 1 Multiple Choice

8875/01

27 September 2013

1 hour

Additional Materials: OMR

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

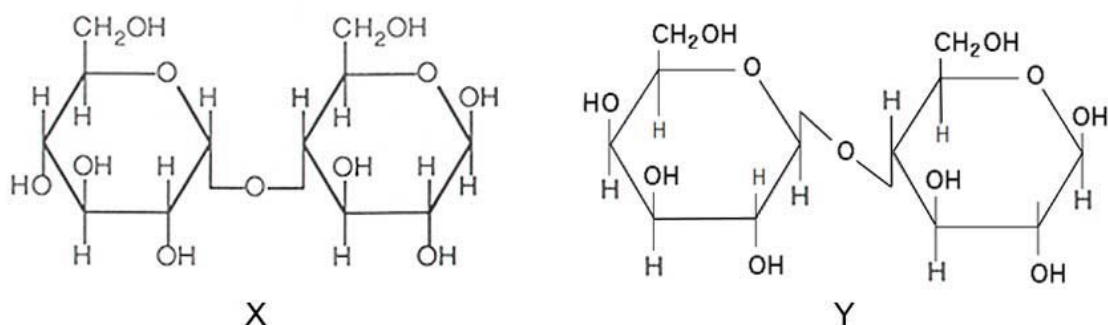
Write your name, civics group and register number in the spaces at the top of this page and all separate paper used.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMR answer sheet.

This paper consists of **17** printed pages

[Turn over]

- 1 The figure below shows two disaccharides, X and Y.



Which statements correctly describe X and Y as shown in the figure?

- 1 Glucose monomer(s) that make up X contains both α -glucose and β -glucose, while the glucose monomer(s) that make up Y contains only β -glucose.
- 2 The glycosidic bond that is formed in X is α -1,2 glycosidic bond.
- 3 The glycosidic bond that is formed in Y is similar to the glycosidic bond that is found in cellulose.
- 4 The general formula of a monomer is $(\text{CH}_2\text{O})_n$.

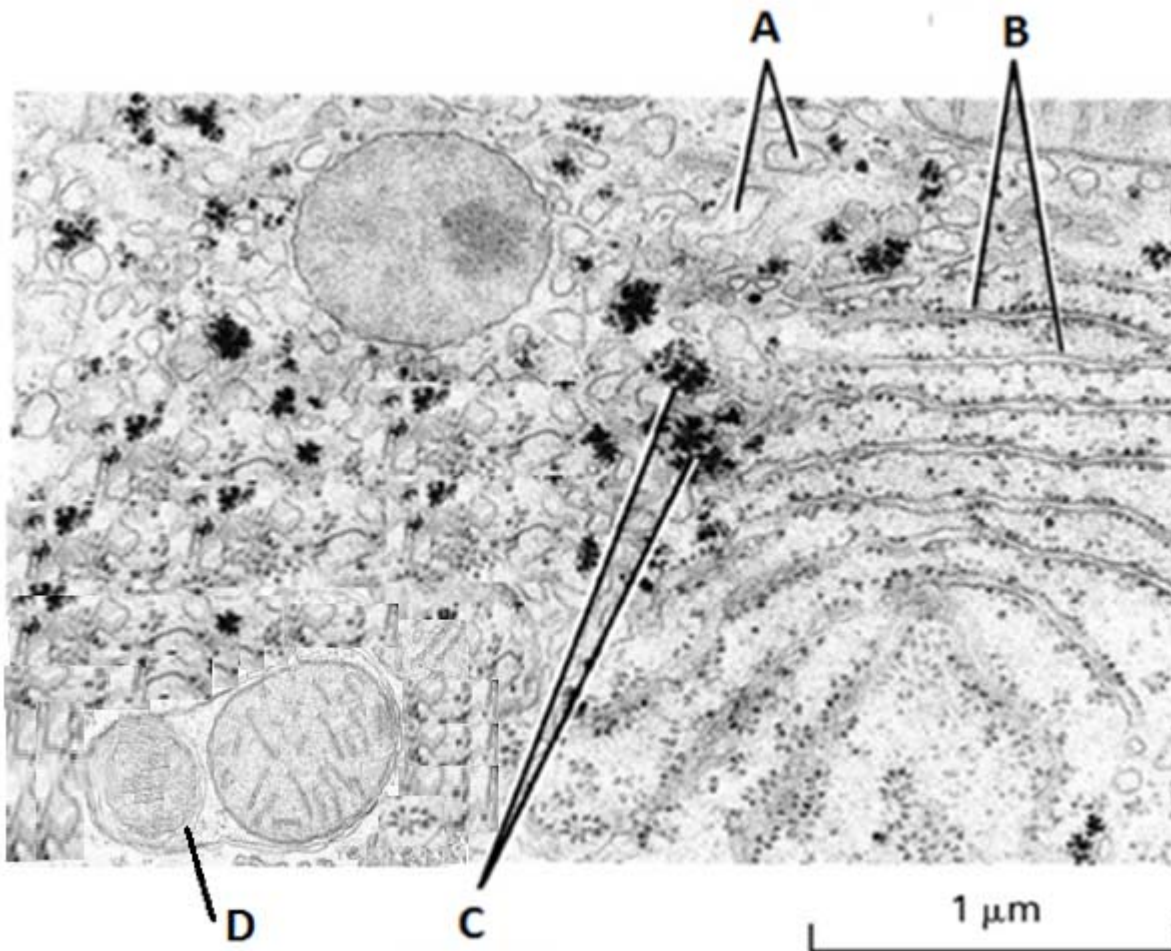
- A 1 and 3 only
- B 2 and 4 only
- C 1, 3 and 4 only
- D 1, 2, 3 and 4

- 2 Which of the following statements are true of both cellulose and protein?

- 1 They may be used as an energy source by certain organisms.
- 2 They are polymers of large molecular mass.
- 3 They act as structural materials in the cell membrane.
- 4 They are synthesized by a series of condensation reactions.

- A 1 and 2 only
- B 2 and 3 only
- C 3 and 4 only
- D 1, 2 and 4 only

3 The electron micrograph below shows a liver cell.



Which statements correctly describe the labelled structures?

- 1 Structure A transports proteins from Structure B to Golgi Apparatus.
 - 2 Proteins enter the lumen of Structure B, where they undergo chemical modifications such as glycosylation.
 - 3 Structure C is starch grain.
 - 4 The process shown in structure D is autolysis.
- A 2 only
- B 1 and 2 only
- C 2 and 3 only
- D 2, 3 and 4 only

- 4 Which of the following statements about the functions of lysosomes are correct?
- 1 They break down hydrogen peroxide, therefore they are also called peroxisomes.
 - 2 They are involved in the phagocytosis of bacteria and autolysis.
 - 3 They secrete β -glucose, which enables plants to construct their cellulose cell walls.
 - 4 They secrete digestive enzymes from the mucosal layer of the stomach.
- A 1 and 3 only
- B 2 only
- C 2 and 4 only
- D 3 and 4 only
- 5 Which one of the following statements about the cell surface membrane is true?
- A It is totally impermeable to water due to the large number of hydrophobic molecules it contains.
- B It is a bilayer, 7 – 10nm thick, containing two layers of intrinsic proteins sandwiched between two layers of phospholipids.
- C It sometimes contains cholesterol which regulates membrane fluidity by preventing it from becoming too fluid at high temperatures or too rigid at low temperatures.
- D Glycolipids and glycoproteins are biological markers which act as antibodies to destroy foreign antigens of the infecting pathogens.

- 6 The table below summarises the results from experiments on the rate of absorption of certain monosaccharides by pieces of mammalian intestine.

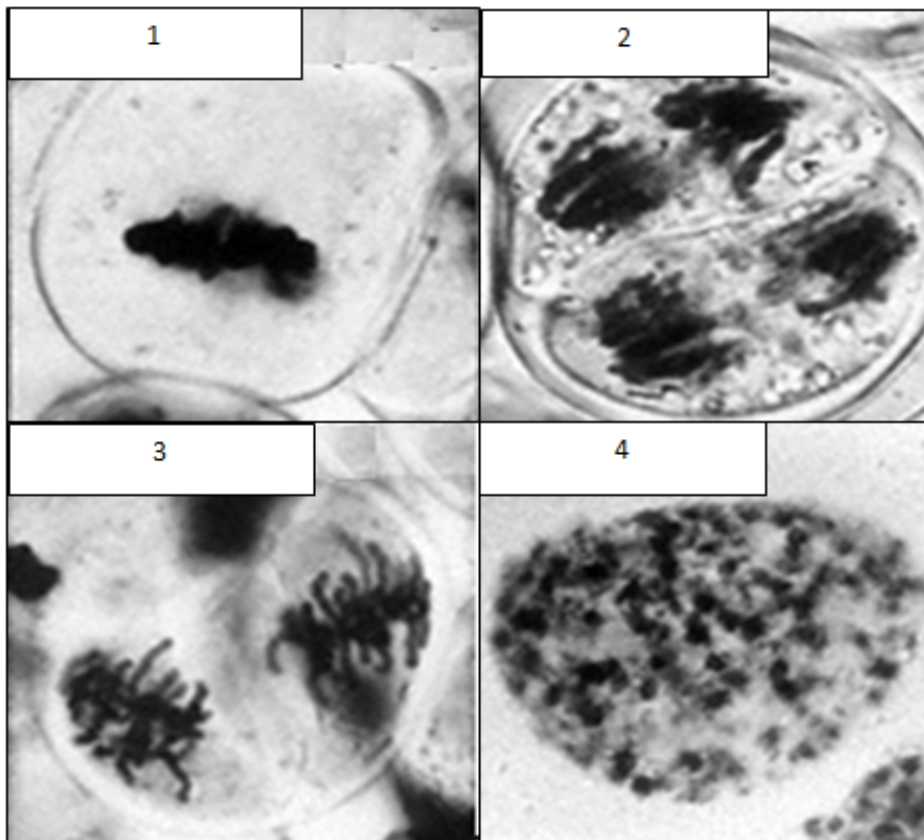
Cyanide inhibits cytochrome c oxidases.

		Relative rate of absorption by intestine (arbitrary units)	
		In physiological saline	In physiological saline with added cyanide
Hexoses	Glucose	1.00	0.33
	Galactose	1.10	0.53
	Fructose	0.43	0.37
Pentoses	Xylose	0.30	0.31
	Arabinose	0.29	0.29

Which of the following statements are consistent with these results?

- 1 Active transport is not involved in the uptake of pentoses.
 - 2 Xylose and arabinose enter by osmosis.
 - 3 Aerobic respiration is necessary for the maximal rate of uptake of glucose.
 - 4 Smaller molecular weight sugars are absorbed more easily.
- A 1 and 3 only
- B 1, 2 and 3 only
- C 1 and 4 only
- D 2 and 4 only

7 The following figure shows nuclear division occurring in a lily plant.



Which of the following shows the correct sequence of events?

- A 1 > 4 > 3 > 2
- B 1 > 3 > 4 > 2
- C 4 > 1 > 3 > 2
- D 4 > 2 > 3 > 1

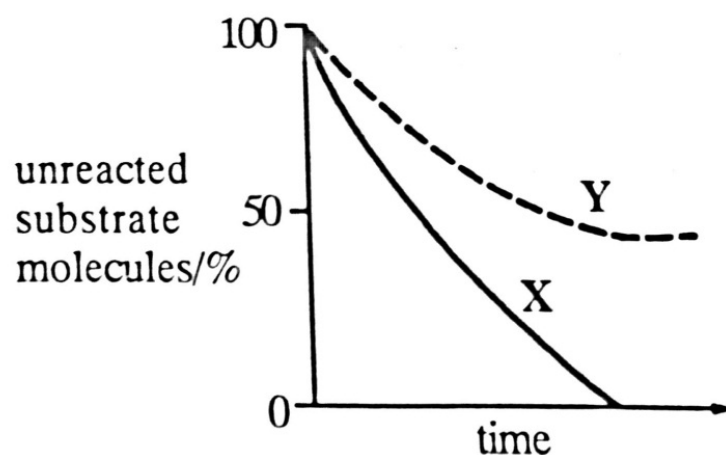
- 8 A biologist counted 2,500 cells from an embryo on a microscope slide and recorded the following data.

Stage	Number of cells
Prophase	125
Metaphase	50
Anaphase	50
Telophase	25
Interphase	2,250

From the table above, it could be reasonably concluded that

- A the cells are undergoing differentiation.
 - B the cells are cancerous.
 - C the cells are dividing randomly.
 - D interphase is the longest stage of nuclear division.
- 9 Which statement about the cell cycle is **not** correct?
- A The G_2 checkpoint checks for DNA damage or mutation.
 - B Rapidly dividing cells have a shorter G_1 phase than non-dividing cells.
 - C Protein synthesis mainly occurs during the G_1 and G_2 phases of the cell cycle.
 - D Dysregulation in cell cycle checkpoints causes chromosomal translocations and cancer.

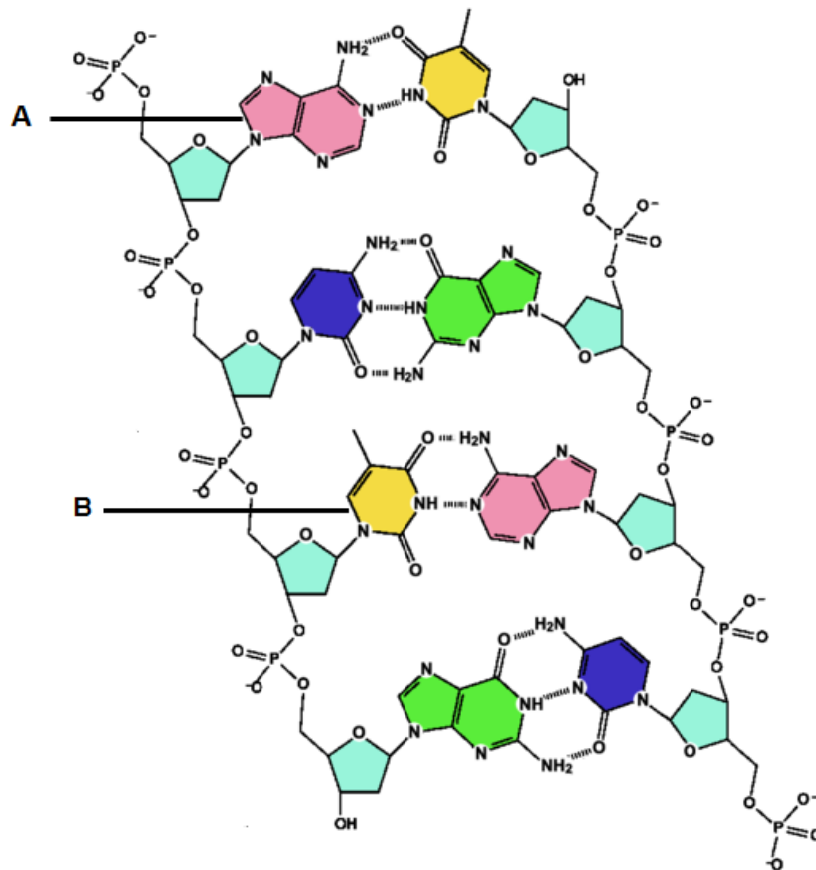
- 10 Curve X represents the course of an enzyme-catalysed reaction under optimum conditions. Curve Y shows the action of the same enzyme on the same substrate but with one alteration to the reaction conditions.



Which of the following factors, operating to a constant extent throughout the experiment, could give the results shown by curve Y?

- A Addition of a competitive inhibitor
- B Addition of an end-product inhibitor
- C An increased substrate concentration
- D A lower temperature

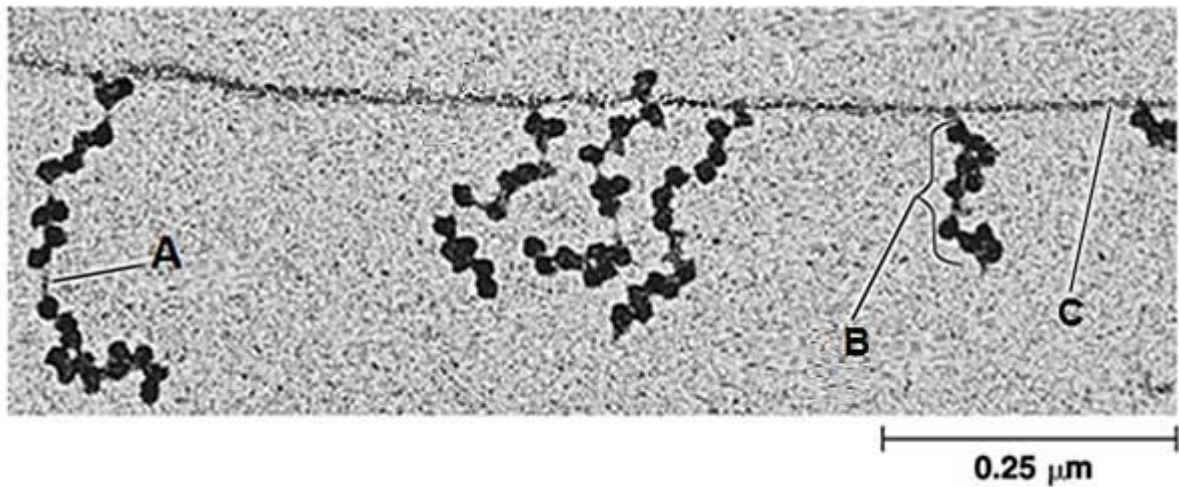
11 The figure below shows a DNA molecule.



Which statements correctly describe the polynucleotide?

- 1 The structure labelled **A** corresponds to a purine, while the structure labelled **B** corresponds to a pyrimidine.
 - 2 The antiparallel nature of DNA double helix aligns the DNA strands so that phosphodiester bonds can form properly between the nitrogenous bases from opposite strands.
 - 3 Distance between adjacent deoxyribonucleotides is $3.4 \text{ \AA}$ and with 10 deoxyribonucleotides per turn. (Note: $10 \text{ \AA} = 1 \text{ nm}$)
 - 4 The wound DNA double helix consists of alternating major grooves and minor grooves along its axis, that is essential for the binding with proteins.
- A 1 only
- B 1 and 2 only
- C 3 and 4 only
- D 1, 3 and 4 only

- 12 The electron micrograph below shows several labelled structures present in a mitochondrion.



Which of the statements below correctly describes the labelled structures?

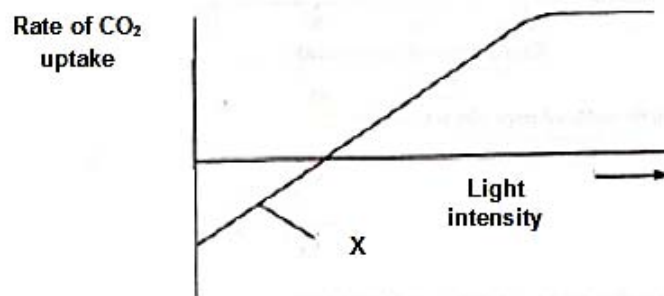
- 1 The structure labelled **A** is the polypeptide chain.
- 2 The structures labelled **B** are polyribosomes. Each ribosome consists of a 50S large subunit and a 30S small subunit.
- 3 The structure labelled **C** is the 3' end of template DNA strand.
- 4 The structure labelled **C** is the 5' end of the mRNA strand.

- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 1, 2 and 4 only

- 13 Which of the following statements correctly describes the type of mutation that gives rise to sickle cell anaemia?

- A** Deletion of a codon resulting in the deletion of amino acid phenylalanine at position 508 of the polypeptide chain.
- B** Deletion of a base pair resulting in the deletion of amino acid phenylalanine at position 508 of the polypeptide chain.
- C** Base pair substitution of thymine by adenine on the non-template strand of DNA causes a change in amino acid from glutamic acid to valine.
- D** Base pair substitution of thymine by adenine on the template strand of DNA causes a change in amino acid from glutamic acid to valine.

- 14 In the graph below, the rate of CO₂ uptake by plant cells is shown to vary with increasing light intensity.



Which of the following is true at point X?

- A The plant is photosynthesising
 - B Respiration equals photosynthesis
 - C CO₂ is a limiting factor
 - D There is not enough light for photosynthesis to have commenced
- 15 Removal of the source of carbon dioxide from photosynthesising chloroplasts results in rapid changes in the concentration of certain chemicals. Which one of the following represents the correct combination of concentration changes?

	<i>ATP</i>	<i>ribulose biphosphate</i>	<i>glycerate-3-phosphate</i>
A	decreases	decreases	increases
B	decreases	increases	no change
C	increases	increases	decreases
D	increases	no change	increases

- 16 Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin. Which additional compound must also be present?

- A acetylcoenzyme A
- B ATP
- C pyruvate
- D reduced NAD

- 17** What happens to most of the reduced NAD molecules in cell metabolism?
- A** They act as oxidising agents in glycolysis.
 - B** They are oxidised in inner mitochondrial membrane for ATP formation.
 - C** They are oxidised in the Calvin cycle.
 - D** They combine with succinic acid as part of Krebs cycle.
- 18** A population's gene pool was found to have remained unaltered for many generations. Which of the following conditions must have existed in the population?
- A** mating had always been random
 - B** geographical isolation had occurred
 - C** inbreeding had been commonly practised
 - D** certain alleles are at selective advantage
- 19** Competition, isolation and selection are thought to be events that take place during the formation of new species. In which order did these events take place in the process of speciation to form 13 species of Darwin's finches?
- A** competition, selection, isolation
 - B** selection, isolation, competition
 - C** isolation, competition, selection
 - D** competition, isolation, selection
- 20** What is the best characteristic for determining whether sexually reproducing flowering plants from separate populations are members of the same species?
- A** ability to cross-fertilize
 - B** ability to cross-pollinate
 - C** occupation of similar niche
 - D** similarity of flower structure

21 Continued self-fertilization in a population of plants heterozygous for a character would normally result in which of the following?

- 1** a decrease in the percentage of recessive homozygotes
- 2** an increase in the percentage of dominant homozygotes
- 3** a decrease in the numbers of recessive homozygotes
- 4** a reduction of heterozygotes by 50% in each generation

- A** 1 and 2
- B** 1 and 3
- C** 2 only
- D** 2 and 4

22 The feature of silky feathers in show fowl is caused by a recessive allele. A pure breeding bird with normal feathers was crossed with a bird with silky feathers and all the offspring were normal. The offspring were then allowed to interbreed.

Which of the following statements would be true about the F₂ generation?

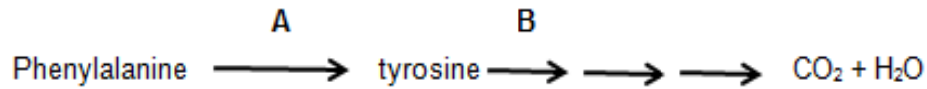
- 1** The expected ratio of normal to silky would be 3:1.
- 2** Half of the F₂ birds would be heterozygous.
- 3** A quarter of the F₂ birds would be homozygous.
- 4** Some of the normal birds would be pure breeding.

- A** 1, 2 and 3
- B** 2, 3 and 4
- C** 1, 2 and 4
- D** 1 and 2 only

23 Colour blindness is determined by a sex-linked recessive allele. A normal male marries a normal female whose father was also normal. The couple then have a colourblind son. From which of his grandparents did the son inherit the allele?

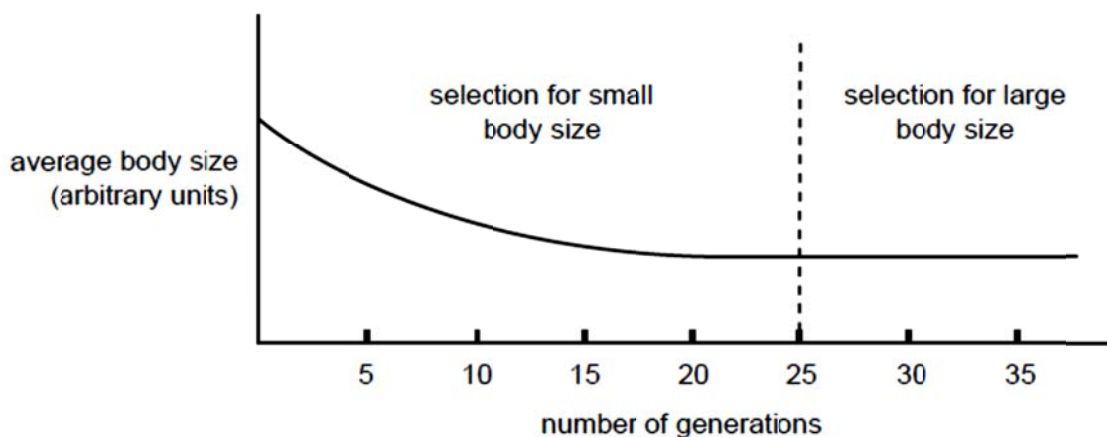
- A** father's mother
- B** father's father
- C** mother's mother
- D** mother's father

- 24 In humans PKU (phenylketonuria) is a disease caused by an enzyme dysfunction at step A in the following simplified reaction sequence, and AKU (alkaptonuria) is due to an enzyme inefficiency in one of the steps summarized as step B.



A person with PKU marries a person with AKU. What are the expected phenotypes for their children? Note: both diseases (PKU and AKU) are not sex linked. Both parents are not heterozygous for both PKU and AKU gene loci.

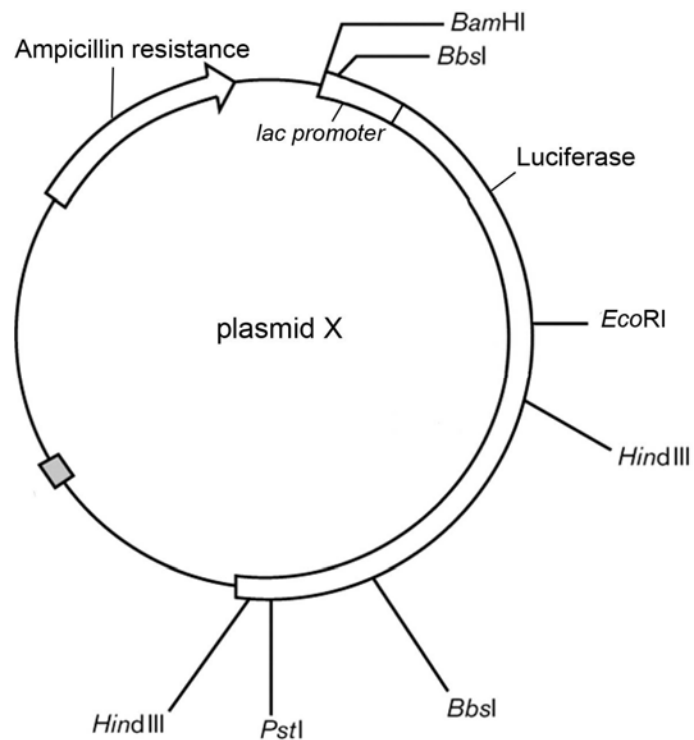
- A All children will be ill.
 - B All children will be normal.
 - C Half of their children will have PKU, but the other half will be normal.
 - D Half of their children will have AKU, but the other half will be normal.
- 25 The graph below shows the results of a breeding experiment with the vinegar fly *Drosophila melanogaster*. In each of the first 25 generations the smallest flies were selected to produce the next generation. After 25 generations the selection was reversed. From generation 25–35 the largest flies were chosen to breed the next generation.



Which of the following can be concluded from the result of the experiment?

- A The flies from generation 25 onwards are phenotypically identical.
- B Body size is inherited in a polygenic manner.
- C Body size is controlled by one gene with more than two alleles.
- D Body size is a sex-linked trait.

26 Plasmid X can serve as a vector for the insertion of genes to be cloned.



Which of the following options will allow the selection of the colonies containing the recombinant form of plasmid X?

	Selection medium	Phenotype of colonies that contain the inserted gene
A	Containing ampicillin and lactose	White colonies
B	Containing ampicillin and luciferase	Colonies that emit light
C	Containing ampicillin, lactose and luciferin	White colonies
D	Containing ampicillin, lactose and luciferin	Colonies that emit light

- 27 A sequence of DNA is to be amplified by the Polymerase Chain Reaction is shown below.



Which option shows suitable primers for amplifying this sequence?

	Primer 1	Primer 2
A	5' – ATTCTCGATCGG – 3'	5' – TCGATATGATCG – 3'
B	5' – CTTCCGATCGAG – 3'	5' – ATTCTCGATCGG – 3'
C	5' – GAAGGCTAGCTC – 3'	5' – GCTAGTATAGCT – 3'
D	5' – CCGATCGAGAAT – 3'	5' – TCGATATGATCG – 3'

- 28 Cloning plants by tissue culture is more advantageous as compared to traditional vegetative propagation methods as
- A The plantlets will require less space to grow to a saleable age.
 - B The plantlets can be grown regardless of seasonal variation.
 - C The plantlets will be genetically identical to the parent plant.
 - D The plantlets will be more resistant to fungi and diseases.
- 29 Recent advances in the field of stem cell research have shown that induced pluripotent stem cells (iPS cells) can be artificially derived from adult somatic cells. iPS cells are mostly similar to natural pluripotent cells. This implies that
- A iPS cells can theoretically differentiate into all cell types.
 - B iPS cells can theoretically differentiate into any of the three germ layers.
 - C iPS cells can theoretically differentiate into gametes.
 - D iPS cells are theoretically capable of transdifferentiation.

30 Which of the following theoretical application(s) of information obtained from the Human Genome Project is/are unethical?

- I. Gene therapies for only certain genetic diseases are developed.
- II. Potential parents undergoing genetic testing and genetic counselling prior to conceiving.
- III. Potential parents choosing embryos for implantation only after ante-natal tests for selected genes.
- IV. Increasing insurance premiums for people who are genetically predisposed to cancer.
- V. A scientist studying phylogeny and early human migration patterns.

- A I and V
- B I and III
- C II and III
- D III and IV

THE END